

NATIONAL UNIVERSITY OF LESOTHO

DEPARTMENT OF PHYSICS & ELECTRONICS

EE3400

LAB 01

EMI 1: Introduction to Electronic Measurements

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Abstract

This lab investigated the operation and practical use of a function generator, oscilloscope and digital multimeter for electronic measurements. Sine, square and triangular waveforms were generated and characterized using peak-to-peak voltage, frequency and RMS voltage. A stepped-down AC supply was also observed. The effect of DC offset and oscilloscope coupling was examined by comparing measurements obtained with the multimeter and oscilloscope. The measured values generally agreed closely with the theoretical RMS relationships. For example, a 5 Vpp sine wave gave 1.76 V on the multimeter and 1.80 V on the oscilloscope, compared with a theoretical value of 1.768 V. A 10 Vpp square wave gave 4.99 V RMS and a 10 Vpp triangular wave gave 2.87 V RMS. The experiment demonstrated the practical relationship between waveform shape, RMS value, DC offset and measurement method.

Introduction

Electronic measurement provides the quantitative information required to design, test and troubleshoot electrical and electronic systems. In this practical, the function generator was used as a controlled source of periodic signals, the oscilloscope was used to visualize voltage as a function of time, and the digital multimeter was used to obtain numerical voltage measurements. These instruments are complementary: the oscilloscope shows waveform shape and timing information, whereas the multimeter provides direct numerical readings.

The practical also demonstrates that RMS voltage depends on waveform shape and that a DC offset must be treated separately from the AC component. This is important in instrumentation because the same peak-to-peak amplitude does not imply the same RMS voltage for different waveforms.

Aim

To understand the operation of the function generator, oscilloscope and digital multimeter, and to use these instruments to generate, observe and measure electrical signals under different waveform, offset and coupling conditions.

Objectives

- Generate and identify sine, square and triangular waveforms.
- Measure peak, peak-to-peak and frequency characteristics using an oscilloscope.
- Measure RMS voltage using both an oscilloscope and a digital multimeter.
- Investigate the effect of DC offset on RMS measurements.
- Compare AC and DC coupling on the oscilloscope.
- Understand basic oscilloscope triggering and its role in obtaining a stable display.

Lab theory

Function generator

A function generator produces controlled electrical waveforms such as sine, square and triangular signals. The main adjustable quantities used in this experiment were amplitude, frequency and DC offset.

Oscilloscope

An oscilloscope displays voltage against time and permits measurement of waveform amplitude, period and frequency. The frequency is related to the period by:

$$f = \frac{1}{\mathrm{T}} \quad (1)$$

DC coupling displays both the AC and DC components of the applied signal. AC coupling blocks the DC component and displays the varying AC component about the reference level. GND coupling disconnects the input signal and displays the zero-voltage reference.

Root-Mean-Square (RMS) Voltage

The RMS value of a periodic voltage is the equivalent DC voltage that would deliver the same average power to a resistive load. The general RMS expression is:

$$V_{\mathrm{rms}} = \sqrt{\frac{1}{\mathrm{T}}\int_0^{\mathrm{T}} v^2(\mathrm{t})\mathrm{dt}} \quad (2)$$

For a symmetrical waveform, the peak voltage and peak-to-peak voltage are related by:

$$V_{\mathrm{p}} = \frac{\mathrm{V_{pp}}}{2} \quad (3)$$

$$V_{\mathrm{pp}} = 2V_{\mathrm{p}} \quad (4)$$

For the waveforms used in the experiment:

Sine wave:

$$V_{\mathrm{rms}} = \frac{\mathrm{V_p}}{\sqrt{2}} = \frac{\mathrm{V_{pp}}}{2\sqrt{2}} \quad (5)$$

Square wave (50% duty cycle):

$$V_{\mathrm{rms}} = V_{\mathrm{p}} = \frac{\mathrm{V_{pp}}}{2} \quad (6)$$

Symmetrical triangular wave:

$$V_{\mathrm{rms}} = \frac{\mathrm{V_p}}{\sqrt{3}} = \frac{\mathrm{V_{pp}}}{2\sqrt{3}} \quad (7)$$

When a DC offset VDC is added to a zero-mean AC waveform, the total true RMS value is:

$$V_{\mathrm{rms}}, total = \sqrt{\mathrm{V_{rms,AC}^2} + \mathrm{V_{DC}^2}} \quad (8)$$

This relationship explains why the RMS value changes when a DC offset is introduced even though the AC waveform itself has not changed shape.

Peak-to-peak voltage represents the complete voltage range from the minimum to the maximum value of the waveform. RMS voltage represents the effective value of the waveform in terms of its power delivered to a resistive load.

Oscilloscope triggering synchronizes the horizontal sweep with a selected point on the waveform, making repetitive waveforms appear stationary. Auto mode continues displaying a trace, Normal mode waits for the trigger condition, and Single mode captures one triggered event and stops.

Apparatus

- Function generator
- Digital oscilloscope with probe
- Digital multimeter
- Step-down transformer for the AC supply measurement
- Connecting leads

Experimental procedure:

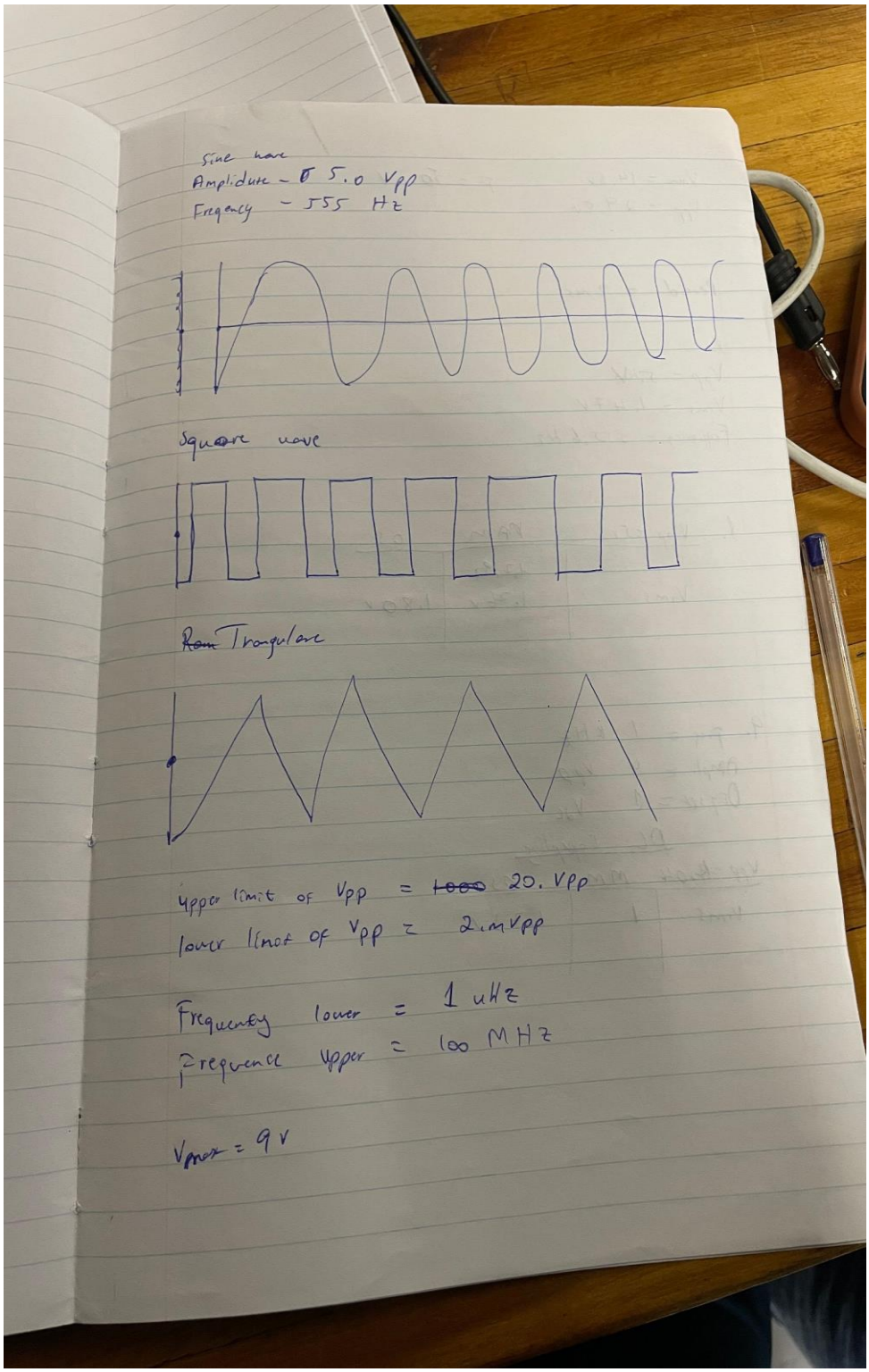
The function generator was connected to the oscilloscope and a stable waveform display was established. Sine, square and triangular signals were then generated, and their shapes were observed on the oscilloscope. The peak-to-peak voltage and frequency of the signals were recorded from the oscilloscope. RMS voltage was measured using the digital multimeter and compared with the corresponding oscilloscope readings. A DC offset was then introduced to a sine wave, and the resulting readings were compared under the relevant coupling and measurement conditions. The stepped-down AC supply was also observed, and its voltage and frequency were recorded. Finally, appropriate oscilloscope triggering and coupling settings were used to obtain stable and interpretable waveforms.

Experimental results:

The following values are taken from the student's handwritten laboratory results. Where a value was not clearly recorded, it has not been invented.

Waveform / condition	**Recorded settings / input**	**Measured result(s)/ VRMS**
Stepped-down AC supply	Vmax =14.8 V; Vpp = 29.8 V; f = 50.0 Hz	Frequency = 50.0 Hz
Sine wave	Vpp =5.00 V; f ≈555–555.6 Hz	DMM=1.76; Oscilloscope=1.80
Sine wave + DC offset	Vpp =4.04 V; VDC =1.00 V; f =1 kHz	Oscilloscope (DC coupling) = 1.73;
Sine wave, no offset	Vpp =10.0 V	DMM =3.52; Oscilloscope = 3.52
Sine wave + 5 V DC offset	Vpp =10.0 V; VDC =5.0 V	DMM = 3.52; Oscilloscope (DC coupling) =6.24
Square wave	Vpp =10.0 V	DMM = 4.99; Oscilloscope = 4.99
Triangular wave	Vpp =10.0 V	DMM = 2.87 V RMS; Oscilloscope = 2.87 V RMS

Experimental result recorded during the laboratory practical:



$V_{max} = 14.8V$ $F = 50.0V$

$V_{pp} = 29.8V$

Period = 2ms

8.

$V_{pp} = 5.12V$

$V_{rms} = 1.47V$

Frequency = 555.6 Hz

8. $V_{pp} = 5V$

	MM	OS
V_{rms}	~~17.92~~ 1.76V	1.80V

9. Freq = 1 kHz

Ampli = 4 V_{pp}

Offset = 1 V_{dc}

DC Coupling

$V_{pp} = 4.04$

	MM	OS
V_{rms}	1	1.73V

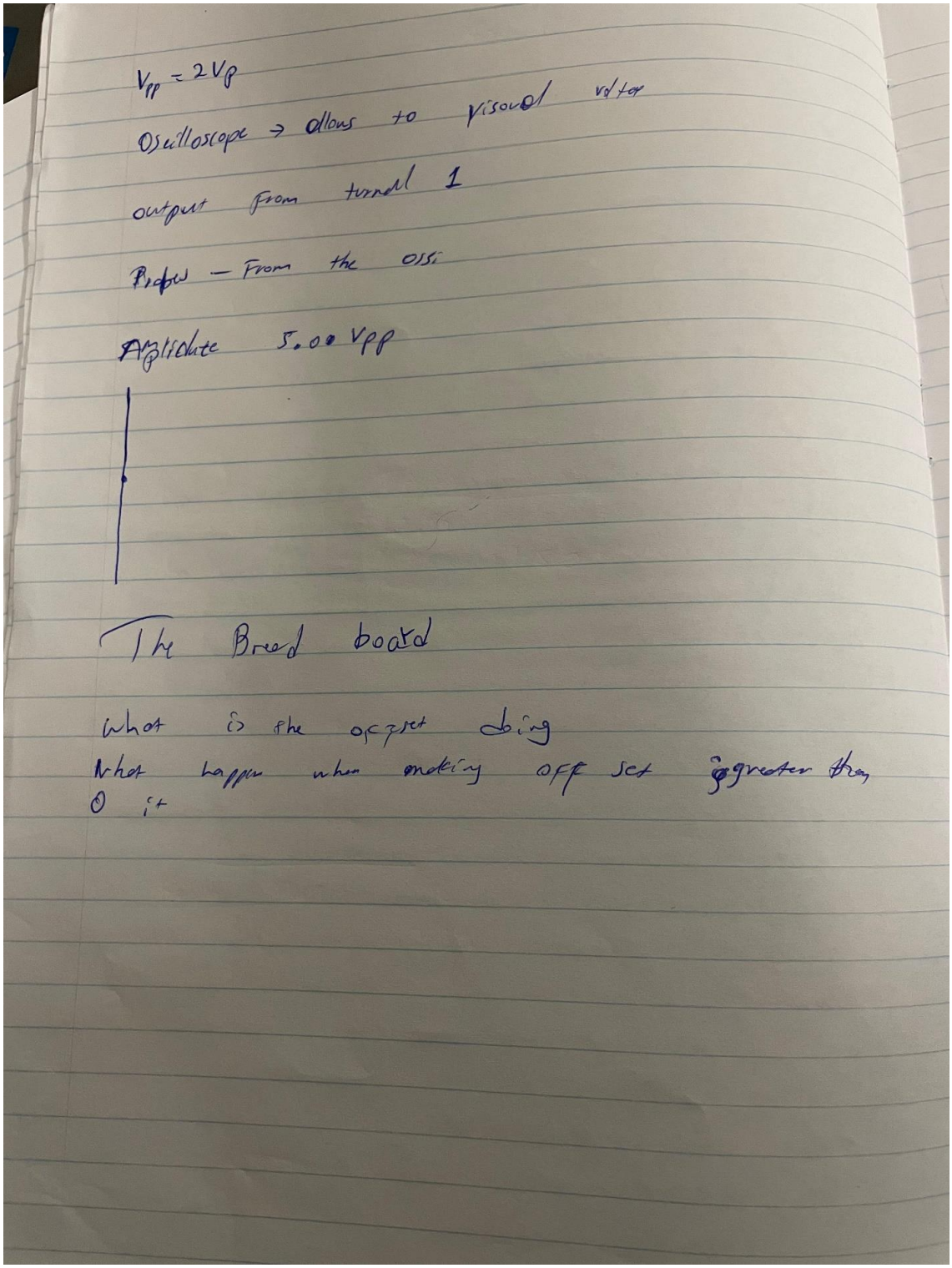
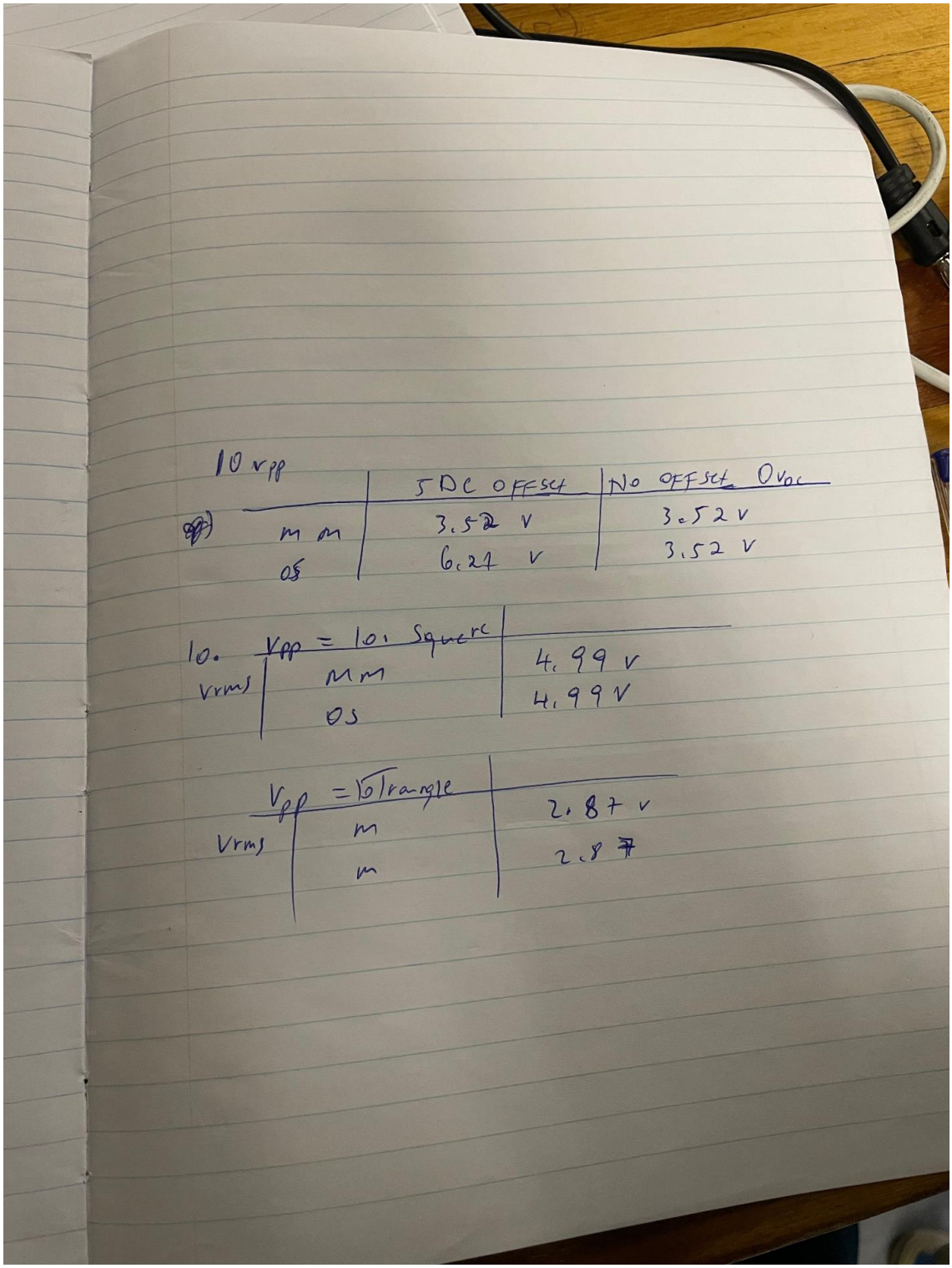


Figure 4: Experimental result recorded during the laboratory practical.



10 Vpp

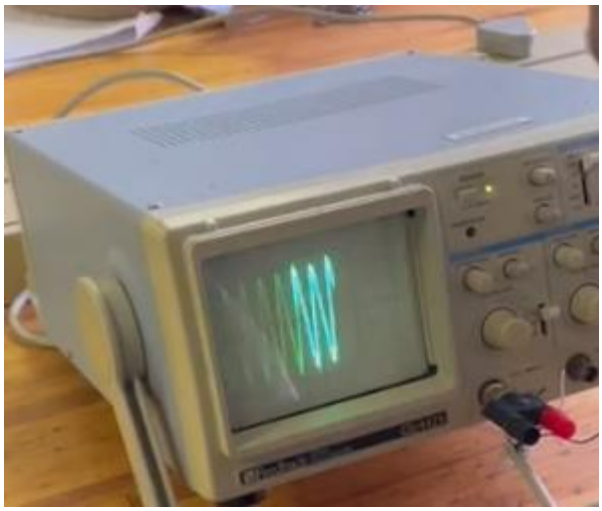
		5 DC OFFSET	No OFFSET 0 Vdc
Vpp	MM	3.52 V	3.52 V
	OS	6.24 V	3.52 V

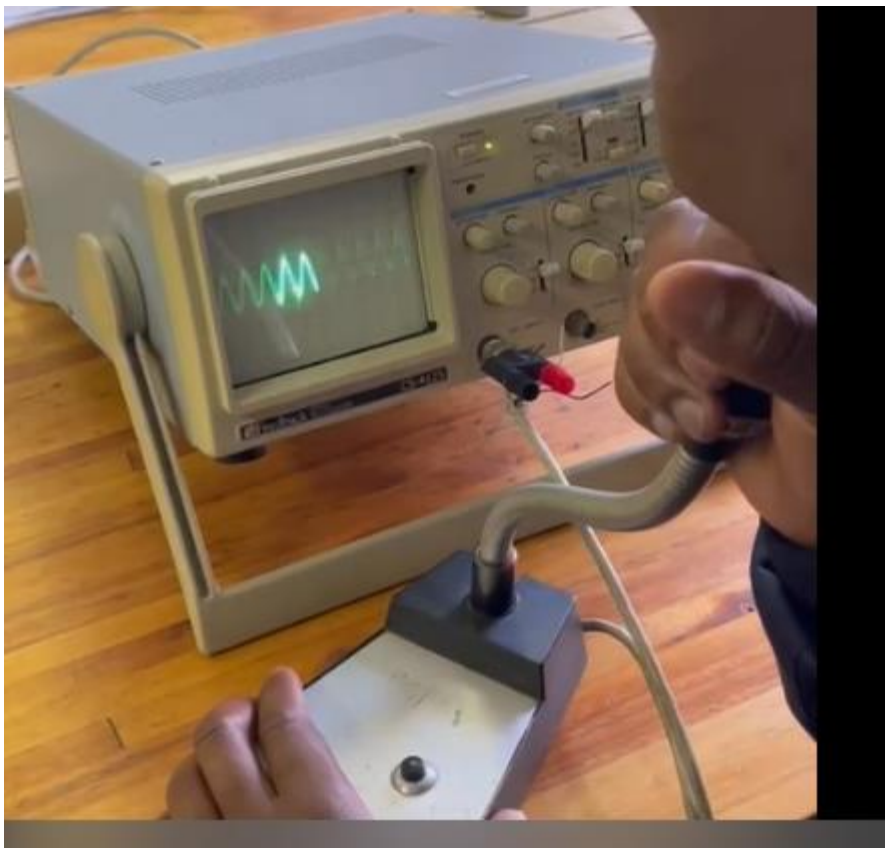
10. Vpp = 10. Square

Vrms	MM	4.99 V
	OS	4.99 V

Vpp = 10 Triangle

Vrms	M	2.87 V
	M	2.87





Function generator amplitude and frequency limits

From Figure 2: the function generator's output amplitude range was recorded as an upper limit of 20 Vpp and a lower limit of 2 mVpp, with a frequency range from 1 µHz to 100 MHz. An open-circuit maximum output of 9 V was also noted.

Calculations

1. 5 Vpp sine wave

$$V_{\text{rms}} = \frac{\text{Vpp}}{2\sqrt{2}} \quad (9)$$

$$V_{\text{rms}}, theory = \frac{5}{2\sqrt{2}}$$

$$= 1.768\, V.$$

$$\%\, Error = \frac{|Vtheoretical - Vmeasured|}{Vtheoretical} \times 100 \quad (10)$$

$$DMM: \frac{|1.76 - 1.768|}{1.768} \times 100 = 0.4525\%$$

$$Oscilloscope: \frac{|1.768 - 1.80|}{1.768} \times 100 = 1.82\%.$$

2. 4.04 Vpp sine wave with 1 V DC offset

$$V_{\text{rms}}, AC = \frac{\text{Vpp}}{2\sqrt{2}} \quad (11)$$

$$V_{\text{rms}}, AC = \frac{4.04}{2\sqrt{2}}$$

$$= 1.428\, V.$$

$$V_{\text{rms}}, total = \sqrt{\text{V}_{\text{rms,AC}}^2 + \text{V}_{\text{DC}}^2} \quad (12)$$

$$= \sqrt{(1.428^2 + 1^2)}$$

$$= 1.744 \text{ V}$$

Using the recorded oscilloscope DC-coupled value of 1.73 V:

$$\%\, Error = \frac{|Vtheoretical - Vmeasured|}{Vtheoretical} \times 100 \quad (13)$$

$$\% \, Error \; = \frac{|1.744 - 1.73|}{1.744} \times 100$$

$$= 0.78\%$$

3. 10 Vpp sine wave with no DC offset

$$V_{\text{rms}} \; = \frac{\text{V}_{\text{pp}}}{2\sqrt{2}} \qquad (14)$$

$$V_{\text{rms}}, theory \; = \frac{10}{2\sqrt{2}}$$

$$= \; 3.536 \, V.$$

$$DMMerror = \frac{|3.536 - 3.52|}{3.536} \times 100$$

$$= 0.44\%$$

$$Oscilloscopeerror = 0.44\%$$

4. 10 Vpp sine wave with 5 V DC offset

$$V_{\text{rms}}, AC \; = \frac{V_{pp}}{2\sqrt{2}} \qquad (15)$$

$$V_{\text{rms}}, AC = \frac{10}{2\sqrt{2}} = \; 3.536 \, V.$$

$$V_{\text{rms}}, total \; = \sqrt{\text{V}_{\text{rms,AC}}^2 + \; \text{V}_{\text{DC}}^2} \qquad (16)$$

$$= \sqrt{(3.536^2 \; + \; 5^2)} \; = \; 6.124 \, V.$$

The recorded oscilloscope DC-coupled value was 6.24 V.

$$\% \, error \; = \frac{|6.124 - 6.24|}{36.124} \times 100$$

$$= \; 1.90\%.$$

5. 10 Vpp square wave

$$V_{\text{rms}} \; = \frac{\text{V}_{\text{pp}}}{2} \qquad (17)$$

$= \frac{10}{2} = 5.00\ V$

Measured = 4.99 V.

$\%\ error\ = \frac{|5.00-4.99|}{5.00} \times 100$

$= \ 0.20\%.$

6. 10 Vpp triangular wave

$V_{\text{rms}}\ = \frac{V_{pp}}{2\sqrt{3}}$ (18)

$V_{\text{rms}}, theory\ = \frac{10}{2\sqrt{3}}$

$= \ 2.887\ V.$

Measured = 2.87 V.

$\%\ error\ = \frac{|2.887\ -2.87|}{2.887} \times 100$

$= \ 0.58\%.$

7. Frequency-period consistency check

$T\ = \frac{1}{f}$ (19)

$$For\ f\ = \ 50.0\ Hz: T\ = \frac{1}{50.0}\ = \ 20\ ms.$$

$$For\ T\ = \ 2\ ms: f\ = \frac{1}{0.002}\ = \ 500\ Hz.$$

Discussion:

The measured RMS values followed the expected dependence on waveform shape. For a 10 Vpp signal, the theoretical RMS values were 3.536 V for a sine wave, 5.00 V for a 50% duty-cycle square wave and 2.887 V for a symmetrical triangular wave. The corresponding recorded measurements were 3.52 V, 4.99 V and 2.87 V, giving percentage errors of only 0.44%, 0.20% and 0.58%, respectively. This close agreement demonstrates that RMS voltage is determined by waveform shape and not by Vpp alone.

The 5 Vpp sine-wave test also showed close agreement: 1.76 V was measured by the multimeter and 1.80 V by the oscilloscope, compared with 1.768 V theoretically. The multimeter therefore differed by 0.44%, while the oscilloscope differed by 1.82%. The difference between the two instruments can arise from instrument resolution, calibration, measurement algorithms and reading uncertainty.

The DC-offset experiment provided the clearest demonstration of the difference between AC and total RMS. For a 10 Vpp sine wave with a 5 V DC offset, the AC component alone has an RMS value of 3.536 V, while the combined true RMS is 6.124 V. The multimeter recorded 3.52 V, whereas the oscilloscope in DC coupling recorded 6.24 V. The latter is close to the calculated total RMS, with a 1.90% error. This result shows that the DC component cannot be ignored when the objective is to determine the total signal power.

The 4.04 Vpp sine wave with a 1 V offset produced 1.73 V on the DC-coupled oscilloscope, compared with 1.744 V calculated from the recorded Vpp and offset. The error was 0.78%, which further supports the RMS relationship for an offset signal.

The stepped-down AC source was recorded at 50.0 Hz and 29.8 Vpp, with a maximum voltage of 14.8 V. The recorded 50 Hz frequency is consistent with the nominal mains frequency stated in the practical.

Discussion Questions:

1. What is the difference between AC, DC and GND coupling?

DC coupling displays the complete input signal, including both AC variation and DC offset. AC coupling blocks the DC component and displays the AC variation about the reference level. GND coupling disconnects the input signal and displays the zero-voltage reference used to position the trace.

2. What is RMS voltage and why is it useful?

RMS voltage is the effective voltage of an AC waveform. It is the value of DC voltage that would produce the same average power in a resistive load. It is useful because different waveforms can have the same Vpp but different power-delivering capability.

Derivation for a sine wave: for v(t) = Vp sin(ωt) over one period T, the mean-square value is:

$$V_{\text{rms}}{}^2 = \frac{1}{T}\int_0^T Vp^2\, sin^2(\omega t)\, dt$$

Using the identity: $sin^2\theta = \frac{1-\cos(2\theta)}{2}$

$$\rightarrow sin^2(wt) = \frac{1-\cos(2wt)}{2}$$

Which implies:

$$[v(t)]^2 = V_p{}^2\,\frac{1-\cos(2wt)}{2} = \frac{V_p{}^2}{2}(1-\cos(2wt))$$

Now we plug into the V_{rms} equation:

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^T \frac{V_p{}^2}{2}(1-\cos(2wt))dt}$$

$$= \sqrt{\frac{V_p{}^2}{2T}\int_0^T (1-\cos(2wt))dt}$$

Evaluating integrals:

$$\int_0^T 1dt = T$$

Second integral:

$$\int_0^T \cos(2wt)\,dt = [\frac{\sin(2wt)}{2w}]_0^T$$

Recall that $2wT = (2T)\frac{2\pi}{T} = 4\pi$, and $\sin(4\pi) = 0$

This implies: $\frac{{V_p}^2}{T}\int_0^T (1 - \cos(2wt))dt = T - 0 = T$

So $V_{rms} = \sqrt{\frac{{V_p}^2}{2T}(T)} = \sqrt{\frac{{V_p}^2}{2}} = \frac{V_p}{\sqrt{2}}$

Since $V_{pp} = 2V_p$ this implies;

$$V_{rms} = \frac{V_p}{\sqrt{2}} = \frac{V_{pp}}{2\sqrt{2}} = [\frac{1}{2\sqrt{2}}]V_{pp}$$

$$V_{\text{rms}} = Vp/\sqrt{2} \approx 0.707\,Vp$$

which matches the multimeter relation given in the laboratory handout.

3. Why can a multimeter and oscilloscope give different readings when DC offset is present?

The instruments may process the signal differently. An AC measurement mode can report the RMS of the alternating component, while an oscilloscope in DC coupling can include both the AC and DC components. Therefore, a signal containing a DC offset can produce different displayed RMS values.

4. What is the purpose of triggering?

Triggering synchronizes the oscilloscope's horizontal sweep with a selected point on the waveform so that repetitive signals appear stationary. Auto Level mode automatically adjusts the trigger level to keep a trace visible even for signals with a drifting or unknown level; Auto mode continues displaying a trace even when a trigger condition is not obtained; Normal mode waits for the trigger condition and

freezes the last trace if none occurs; Single mode captures one triggered event and stops.

5. What are DC and AC trigger coupling?

DC trigger coupling allows the complete signal, including its DC level, to participate in trigger detection. AC trigger coupling blocks the DC component in the trigger path so that the trigger responds to the varying AC portion of the signal.

6. What scope settings (trigger, V/div and time/div) would capture a transient consisting of two 1 V pulses, 100 ns wide, separated by 10 ms?

Because the pulses do not repeat, Single (single-shot) trigger mode is required, with the trigger source set to the same channel, DC trigger coupling (so the trigger responds to the true pulse level rather than an AC-coupled, filtered version of it), a rising-edge slope, and a trigger level of roughly 0.5 V (half the 1 V pulse height) to reject noise while still catching the pulse reliably. For V/div, a setting of about 0.2 V/div displays the 1 V pulses across five divisions with headroom above and below. For time/div, two regimes are useful: about 20 ns/div resolves the shape of a single 100 ns pulse across roughly five divisions, while a slower setting of about 1 ms/div fits the full 10 ms separation within a 10-division screen, showing both pulses as narrow spikes 10 divisions apart at the cost of pulse-shape detail. In practice the pulse shape and the pulse spacing cannot both be resolved on one screen at once, so the faster timebase is used to verify each pulse individually, and the slower timebase (or a scope with delayed/zoomed timebase) is used to confirm the 10 ms spacing.

Sources of error

- finite measurement resolution.
- Different RMS calculation or measurement methods used by the multimeter and oscilloscope.
- Reading uncertainty when setting or reading Vpp and DC offset.
- Function-generator output tolerance and possible loading by the measuring instrument.
- Probe and connecting-lead effects.
- The recorded inconsistency between 50.0 Hz and 2 ms for the stepped-down AC measurement.

Percentage-error summary

Waveform / condition	Theoretical RMS (V)	Measured RMS (V)	Percentage error
5 Vpp sine DMM	1.768	1.76	0.44%
5 Vpp sine Oscilloscope	1.768	1.80	1.82%
4.04 Vpp sine + 1 V DC Oscilloscope	1.744	1.73	0.78%
10 Vpp sine DMM	3.536	3.52	0.44%
10 Vpp sine Oscilloscope	3.536	3.52	0.44%
10 Vpp sine + 5 V DC Oscilloscope	6.124	6.24	1.90%
10 Vpp square	5.000	4.99	0.20%
10 Vpp triangular	2.887	2.87	0.58%

Conclusion

The practical successfully demonstrated the basic operation and complementary roles of the function generator, oscilloscope and digital multimeter. The measured RMS values agreed closely with theory for sine, square and triangular waveforms. For 10 Vpp signals, the measured sine-, square- and triangle-wave RMS values were 3.52 V, 4.99 V and 2.87 V, respectively, compared with theoretical values of 3.536 V, 5.00 V and 2.887 V. The corresponding percentage errors were 0.44%, 0.20% and 0.58%. The DC-offset tests further showed that the AC component and total signal RMS are different quantities: a 10 Vpp sine wave with a 5 V offset produced a calculated total RMS of 6.124 V and a recorded DC-coupled oscilloscope value of 6.24 V, giving a 1.90% error. The experiment therefore established the practical importance of waveform shape, RMS measurement, coupling selection and correct interpretation of instrument readings in electronic instrumentation.

References

- National University of Lesotho, EE3400 Electronic Measurements and Instrumentation, Lab 01: Introduction to Electronic Measurements, laboratory handout, 2026.
- National University of Lesotho, EE3400 Lecture Slides.
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- Doebelin, E. O., Measurement Systems: Application and Design, 5th ed., McGraw-Hill.